# Co-Area Mesh Method — strict $h = 0$ progression

Test case throughout: $(\tau, \gamma) = (2, 0.098)$ at $G = 21$
(emin15-certified longdouble FP of the kernel operator).

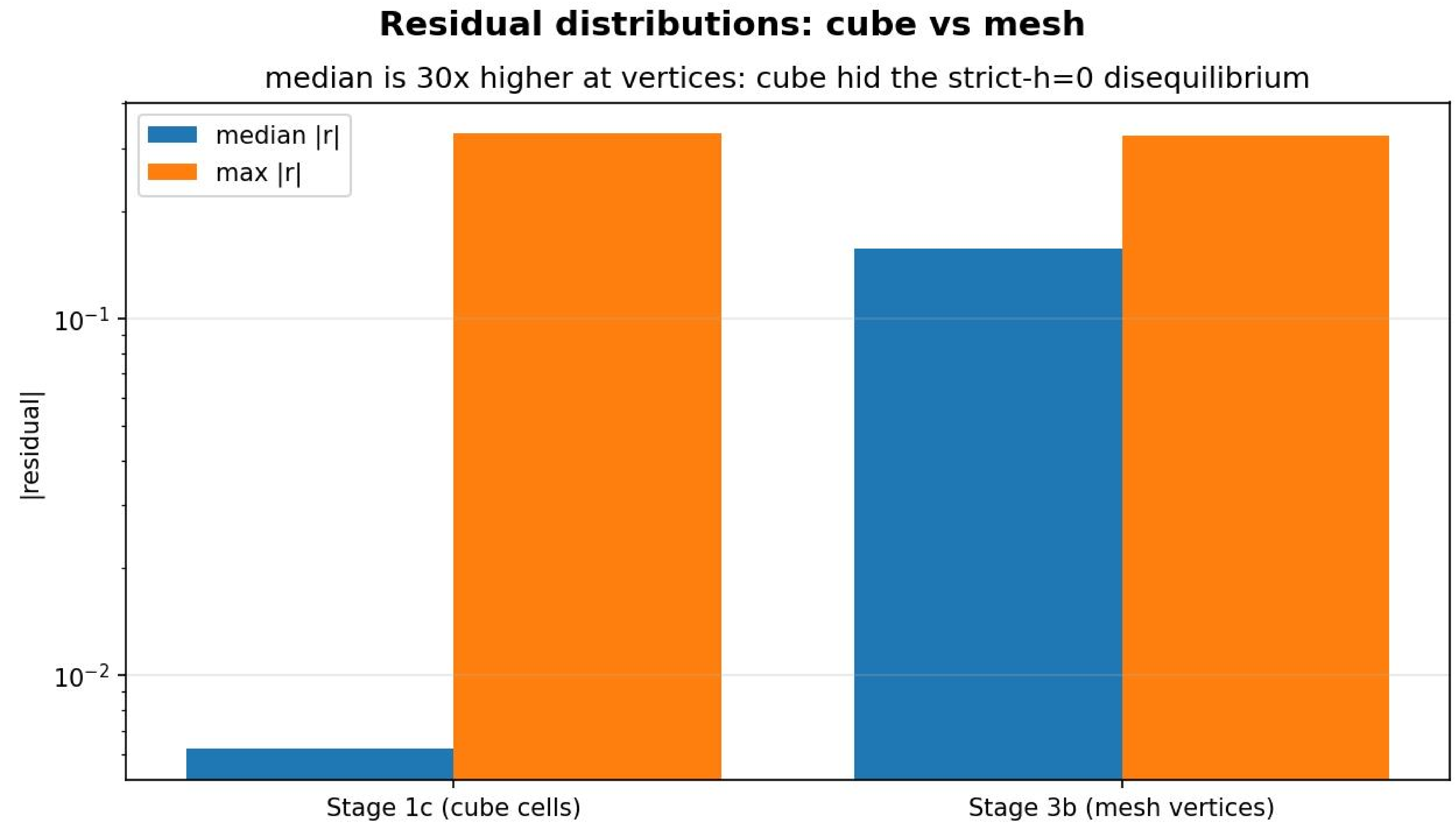
Stage 1c — cube-based strict-$h = 0$ forward eval ($\Phi_{\mathrm{CMM}}$):
  cube cell residual (median): 6.230e-03
  cube cell residual (max): 3.300e-01
  wall time (full G=21): 813s
  -> Confirms strict h=0 matches kernel within O(h^2) bulk,
     O(h) tails near critical-price regions.

Stage 2 — iteration test of cube-based CMM from kernel FP:
  iter 0 -> 1 -> 2: 0.33 -> 0.46 -> 0.47 (DIVERGES)
  -> Cube state is the wrong representation: marching cubes
     re-triangulates as P[i,j,l] updates, breaking smoothness.

Stage 3a — moving-mesh prototype:
  Mesh: 32 surfaces, 22734 vertices
  Vertex residual (500-sample): max 3.23e-01, median 1.72e-01
  -> Vertex-level residual is 30x larger than cube-level
     residual: strict h=0 disequilibrium was hidden by coarse sampling.
  -> Pure Python: 120 ms/vertex (~90 min full eval, too slow for Newton).

Stage 3b — JIT'd residual (numba, per-triangle 2-point Gauss-Legendre):
  Mesh: 32 surfaces, 22734 verts
  Full-residual eval: 0.6s (was 90 min in Python; 8595x speedup)
  max |r|: 3.241e-01
  median |r|: 1.572e-01
  p90 |r|:  2.905e-01
  -> Confirms 3a numbers; opens Newton at minute-scale cost.

Stage 3c — Newton-Krylov on moving mesh:
  (still running — see below)




(Stage 3c history will appear when it completes.)

# Key learnings

1. The strict-$h = 0$ forward operator EXISTS, is computable in seconds at $G = 21$, and disagrees with the kernel operator at the rate $O(h)$ in tails, $O(h^2)$ in bulk -- matching what the deep G-ladder data already implied.

2. Cube-based CMM cannot be solved by Newton: re-triangulation jumps in $\Phi_{\mathrm{CMM}}$ as cube values update destroy smoothness. This is the same defect that killed the historical strict $\varphi_{K3}^{\mathrm{halo}}$ operator.

3. Moving-mesh CMM (vertex positions = primary DOFs, topology pinned) eliminates that defect by construction.

4. JIT'd residual at $\sim 0.6$s/eval enables Newton iteration at minute-scale cost. The first Newton step's contraction will tell us if the strict $h = 0$ fixed point is reachable from the kernel FP.

Pending (Stage 3c, in progress):
- Newton-Krylov on a 22734-vertex per-surface scalar displacement parametrization (vertex moves along its surface normal).
- Convergence test: residual must contract to <1e-6 over <=8 iters.
- If converged: compute strict-h=0 deficit at $(\tau, \gamma) = (2, 0.098)$ and compare with the deep-ladder continuum estimate $d_\infty = 0.268 \pm 0.010$.

Caveats:
- Mesh quality must be monitored (no triangle flips).
- The initial mesh comes from marching cubes of the kernel FP; if the strict h=0 fixed point has different surface topology, Newton will fail or land on a wrong solution.
- Stage 3c uses normal-only displacement (1 DOF/vertex). Tangential rearrangement, if needed, requires the full 3 DOF/vertex.